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EXP - 11

TITLE - Multi - Thread Application

```
In [26]: import threading, random, time

number = 0

e1 = threading.Event()
e2 = threading.Event()

def generate():

    global number

    while True:

        number = random.randint(1, 10)

        print("Generated number:", number)

        e1.set()

        time.sleep(1)

def square():

    while True:

        e1.wait()

        print("Square of", number, "=", number**2)

        e1.clear()

        e2.set()

def cube():

    while True:

        e2.wait()

        print("Cube of", number, "=", number**3)

        print("=====")

        e2.clear()
```

```
t1 = threading.Thread(target=generate)

t2 = threading.Thread(target=square)

t3 = threading.Thread(target=cube)

t1.start()

t2.start()

t3.start()
```

Generated number: 5

Square of 5 = 25

Cube of 5 = 125

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Generated number: 3

Square of 3 = 9

Cube of 3 = 27

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Generated number: 3

Square of 3 = 9

Cube of 3 = 27

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Generated number: 6

Square of 6 = 36

Cube of 6 = 216

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In []: